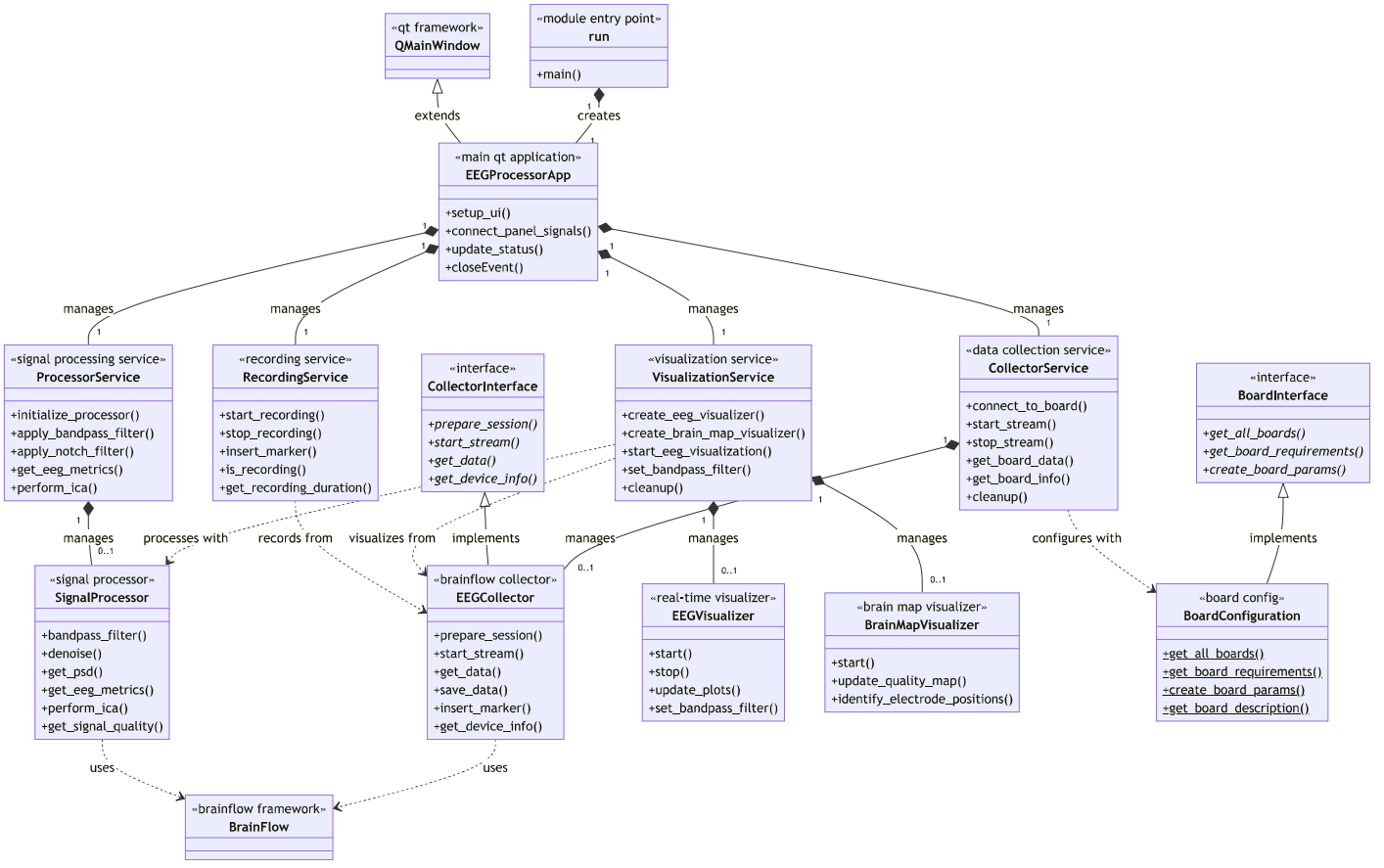

«qt framework»
QMainWindow
«module entry point»
run
+main()
extends
creates
«main qt application»
EEGProcessorApp
+setup_ui()
+connect_panel_signals()
+update_status()
+closeEvent()
manages
manages
manages
manages
«signal processing service»
ProcessorService
+initialize_processor()
+apply_bandpass_filter()
+apply_notch_filter()
+get_eeg_metrics()
+perform_ica()
«recording service»
RecordingService
+start_recording()
+stop_recording()
+insert_marker()
+is_recording()
+get_recording_duration()
«interface»
CollectorInterface
+prepare_session()
+start_stream()
+get_data()
+get_device_info()
«visualization service»
VisualizationService
+create_eeg_visualizer()
+create_brain_map_visualizer()
+start_eeg_visualization()
+set_bandpass_filter()
+cleanup()
«data collection service»
CollectorService
+connect_to_board()
+start_stream()
+stop_stream()
+get_board_data()
+get_board_info()
+cleanup()
«interface»
BoardInterface
+get_all_boards()
+get_board_requirements()
+create_board_params()
manages
processes with
records from
visualizes from
implements
manages
manages
manages
configures with
implements
«signal processor»
SignalProcessor
+bandpass_filter()
+denoise()
+get_psd()
+get_eeg_metrics()
+perform_ica()
+get_signal_quality()
«brainflow collector»
EEGCollector
+prepare_session()
+start_stream()
+get_data()
+save_data()
+insert_marker()
+get_device_info()
«real-time visualizer»
EEGVisualizer
+start()
+stop()
+update_plots()
+set_bandpass_filter()
«brain map visualizer»
BrainMapVisualizer
+start()
+update_quality_map()
+identify_electrode_positions()
«board config»
BoardConfiguration
+get_all_boards()
+get_board_requirements()
+create_board_params()
+get_board_description()
uses
uses
«brainflow framework»
BrainFlow